Crop Disease Detection System Using Image Processing and Deep Learning

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***Abstract*—Crop disease is one of the most critical threats to agricultural productivity, with 30% to 40% of crops affected annually. This paper proposes a feature fusion framework for automatic crop disease detection using leaf images from the PlantVillage dataset, which contains over 39,000 images across 38 disease classes. The framework combines three complementary feature extraction techniques: Gray-Level Co-occurrence Matrix (GLCM) for texture analysis, Histogram of Oriented Gradients (HOG) for structural shape features, and MobileNetV2 for deep semantic features. The fused 2,852-dimensional feature vector is classified using a Light Gradient Boosting Machine (LightGBM) classifier. Evaluated on 40,612 test samples across 17 disease classes, the proposed system achieves an overall accuracy of 96.17% with an average inference time of 0.42 seconds per image, outperforming individual feature methods and existing hybrid models.**

***Index Terms*—crop disease detection, feature fusion, GLCM, HOG, MobileNetV2, LightGBM, deep learning, PlantVillage**

I. Introduction

Agriculture is identified as the backbone of any country, and in India it has historically been a primary source of income for 58% of the population. India is the world's 6th largest market for food and grocery, making agriculture one of the most crucial sectors of the economy [1]. Crops play a fundamental role in food security and economic stability, yet 30% to 40% of crop yield is lost annually due to plant diseases, posing a serious threat to global food security.

Plant diseases exert a significant negative influence on farmers' livelihoods and contribute to global food insecurity. The health of a plant's leaves is often the earliest visible indicator of disease, making leaf-based diagnosis an important approach for disease monitoring. Common crops such as potato, tomato, and pepper are susceptible to diseases that cause heavy financial losses if not detected early.

Conventional disease identification relies on agriculture experts conducting on-site assessments or experienced farmers making visual judgments. This approach is subjective, time-consuming, labor-intensive, and prone to error, especially among less experienced farmers. Incorrect disease identification can lead to inappropriate chemical use, environmental pollution, and unnecessary economic losses. To address these limitations, automated image-based plant disease detection has emerged as an active and important research topic.

This paper proposes a multi-feature fusion framework that combines GLCM texture descriptors, HOG structural features, and MobileNetV2 deep features into a unified 2,852-dimensional vector classified by LightGBM, deployed as a React + FastAPI web application for real-time field use.

This paper makes the following contributions:

- A hybrid feature fusion pipeline combining GLCM, HOG, and MobileNetV2 into a unified 2,852-dimensional representation, capturing complementary visual properties that no single method provides alone.
- An HSV-based background segmentation step that isolates the leaf region from complex field backgrounds, reducing noise and improving feature quality.
- A comprehensive ablation study across six feature combinations demonstrating that full three-way fusion achieves 96.17% accuracy, a 10.40 percentage point gain over MobileNetV2 alone.
- A full-stack web application (React + FastAPI) with average inference latency of 0.42 seconds per image, making the system practically deployable for farmers and extension workers.
- Evaluation on 40,612 test samples across 17 disease classes with per-class precision, recall, and F1-score analysis.

II. Literature Review

Significant research has been conducted in the domain of crop disease detection using both machine learning and deep learning techniques. This section reviews key contributions. Table I summarizes the methods discussed.

A. YOLO-Based Real-Time Detection

Morbekar et al. (2020) proposed a real-time crop disease detection system based on the YOLO architecture [10]. Unlike traditional methods, YOLO performs detection and classification simultaneously on both local and global image regions. The model was trained on labeled leaf images and detects diseased regions by predicting bounding boxes with class labels, making it suitable for field deployment.

B. GLCM with CNN Hybrid Model

Meena et al. (2024) proposed a hybrid model combining GLCM-based texture features with a Convolutional Neural Network (CNN) [11]. GLCM captures statistical texture patterns from leaf images, while the CNN extracts high-level spatial features. This combined approach achieved a classification efficiency of 87%, outperforming several existing models.

C. HOG with FK-NN and CNN

Gama et al. (2025) created a system using HOG for feature extraction followed by classification using both a Fuzzy K-Nearest Neighbor (FK-NN) model and a CNN [12]. FK-NN outperformed CNN with an accuracy of 86.26%; however, its slower prediction speed limits real-time applicability.

D. Transfer Learning Using MobileNetV2

Jesu Priya et al. applied MobileNetV2 with transfer learning for apple leaf disease detection. The model utilizes pretrained ImageNet weights with additional classification layers, achieving approximately 97% accuracy. However, its scope was limited to apple crops and lacked spatial disease localization.

E. LightGBM-Based Machine Learning

Zhang et al. presented a model combining LightGBM with Logistic Regression for classification, employing SMOTE to handle class imbalance. The model achieved approximately 94.7% accuracy with strong AUC values while minimizing manual feature engineering.

F. AlexNet-Based Deep Learning

Researchers using AlexNet with brightness enhancement and noise removal preprocessing on the PlantVillage dataset achieved approximately 94.5% accuracy. While effective, AlexNet demands considerable computational resources compared to lighter architectures.

TABLE I
SUMMARY OF RELATED WORK IN CROP DISEASE DETECTION

Reference	Method	Accuracy	Limitation
Morbekar et al. [10]	YOLO	—	No class metrics
Meena et al. [11]	GLCM + CNN	87%	Limited crops
Gama et al. [12]	HOG + FK-NN	86.26%	Slow inference
Jesu Priya et al.	MobileNetV2 TL	∼97%	Apple only
Zhang et al.	LightGBM+LR	94.7%	Complex pipeline
AlexNet study	AlexNet	94.5%	High compute
Proposed	**GLCM+HOG+ MobileNetV2**	**96.17%**	—

III. METHODOLOGY

A. Dataset Preparation

The PlantVillage dataset [7], available on Kaggle, was used in this study. It comprises more than 39,000 high-resolution images across 38 classes of diseased and healthy leaves from various plant species. From each of the 38 classes, 200 images were sampled for experiments. Table II summarises the dataset configuration.

TABLE II
DATASET CONFIGURATION

Parameter	Value
Total classes	38
Images per class (sampled)	200
Train / Test split	80% / 20%
Sampling strategy	Stratified
Image source	PlantVillage [7]
Input size (GLCM/HOG)	256×256 px
Input size (MobileNetV2)	224×224 px

B. System Architecture Overview

The overall pipeline is shown in Fig. 1. The system follows four main stages: (1) image acquisition and HSV-based background segmentation, (2) parallel multi-modal feature extraction using GLCM, HOG, and MobileNetV2, (3) feature fusion and normalization into a 2,852-dimensional vector, and (4) LightGBM classification with a JSON response returned to the React web frontend.

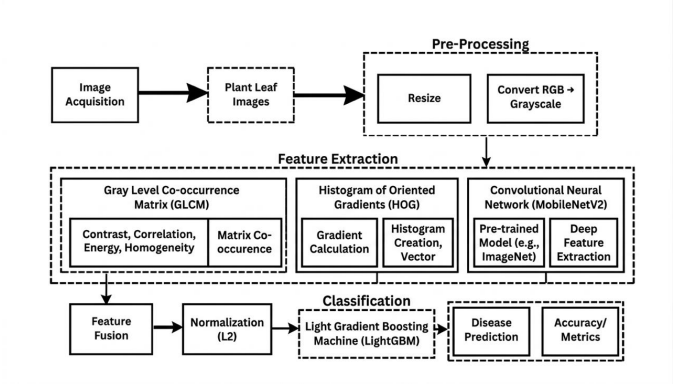


Fig. 1. System architecture of the proposed CropCare pipeline. The hybrid ML pipeline receives a leaf image, performs HSV segmentation, extracts GLCM (4-dim), HOG (1,568-dim), and MobileNetV2 (1,280-dim) features, fuses them into a 2,852-dimensional vector, and classifies with LightGBM.

C. Preprocessing: HSV-Based Background Segmentation

Raw leaf photographs captured in field conditions contain cluttered backgrounds and variable lighting. To isolate the leaf

region, HSV color space segmentation is applied. The image is converted from BGR to HSV, and a green-range mask retains only the leaf pixels. Fig. 2 illustrates the effect on a sample leaf image.

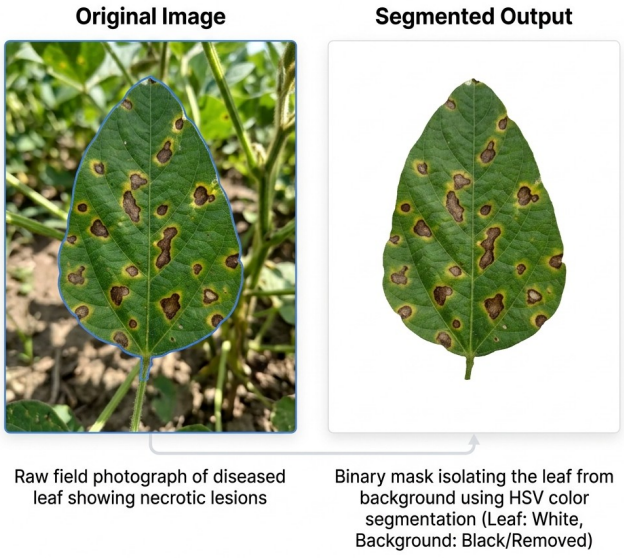


Fig. 2. HSV-based background removal. Left: original leaf photograph with cluttered background. Right: segmented leaf region after HSV masking, retaining only the leaf pixels for feature extraction.

D. Feature Extraction

Crop disease images exhibit variation in texture, shape, and appearance. Relying on a single feature type may lead to incomplete representation. Three complementary feature extraction methods were employed. Table III compares them.

TABLE III
COMPARISON OF FEATURE EXTRACTION METHODS

Property	GLCM	HOG	MobileNetV2
Type	Statistical	Structural	Deep CNN
Extraction	Manual	Manual	Automated
Output dims	4	1,568	1,280
Captures	Texture	Shape/Edges	Semantics
Compute cost	Low	Moderate	High

1) Deep Feature Extraction using MobileNetV2: A pretrained MobileNetV2 convolutional neural network was used as a deep feature extractor. The Global Average Pooling (GAP) layer extracts a 1,280-dimensional deep feature vector per image. Fig. 3 illustrates the MobileNetV2 feature extraction flow.

2) Texture Feature Extraction using GLCM: Gray-Level Co-occurrence Matrix (GLCM) features were extracted to capture texture patterns [1]. Each image was converted to grayscale and resized to 256×256 pixels. A normalized co-occurrence matrix was computed with distance 1 at angle $0°$. Five statistical measures were derived:

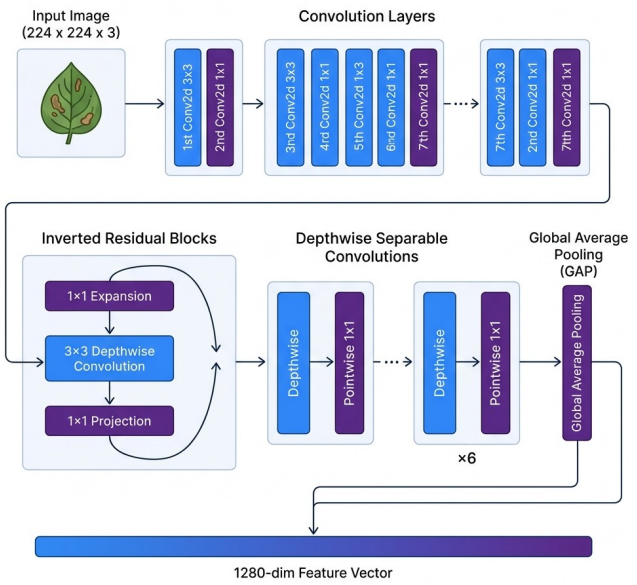


Fig. 3. MobileNetV2 deep feature extraction. The pretrained network processes a 224×224 leaf image through depthwise separable convolution blocks. The GAP layer outputs a 1,280-dimensional feature vector representing high-level semantic disease patterns.

$$\text{Contrast} = \sum_{i,j=0}^{L-1} P_{i,j}(i-j)^2 \tag{1}$$

$$\text{Dissimilarity} = \sum_{i,j=0}^{L-1} P_{i,j}|i-j| \tag{2}$$

$$\text{Homogeneity} = \sum_{i,j=0}^{L-1} \frac{P_{i,j}}{1+(i-j)^2} \tag{3}$$

$$\text{ASM} = \sum_{i,j=0}^{L-1} P_{i,j}^2 \tag{4}$$

$$\text{Energy} = \sqrt{\text{ASM}} \tag{5}$$

where $P_{i,j}$ is the normalized co-occurrence probability and L is the number of gray levels.

3) Structural Feature Extraction using HOG: Histogram of Oriented Gradients (HOG) captures edge directions and structural shape information [3]. The gradient orientation $\theta_{x,y}$ at each pixel is:

$$\theta_{x,y} = \tan^{-1} \frac{L(x,y+1) - L(x,y-1)}{L(x+1,y) - L(x-1,y)} \tag{6}$$

where L is the pixel intensity. HOG was computed using 9 orientations, 8×8 pixels per cell, and 2×2 cells per block, yielding 1,568 dimensions. Fig. 4 shows a HOG visualization on a diseased leaf.

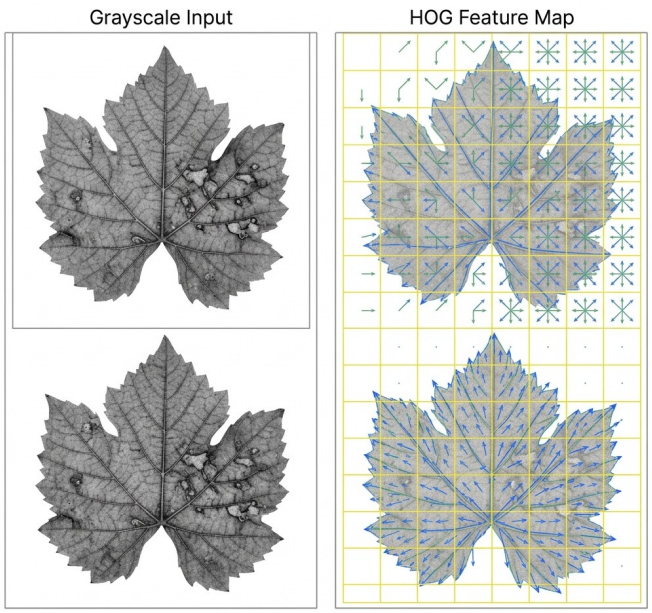


Fig. 4. HOG feature visualization on a diseased tomato leaf. Left: original grayscale image. Right: HOG gradient orientation map showing dominant edge directions that characterize disease lesion boundaries and leaf venation structure.

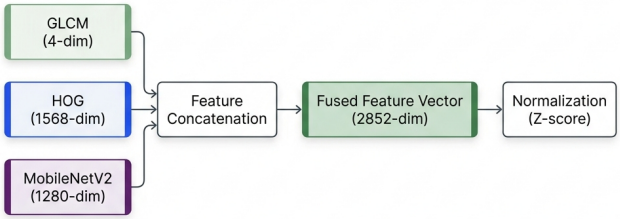


Fig. 5. Feature fusion diagram. GLCM (4-dim), HOG (1,568-dim), and MobileNetV2 GAP (1,280-dim) vectors are concatenated into a 2,852-dimensional feature vector, which is L2-normalized before LightGBM classification.

E. Feature Fusion

The GLCM (4-dim), HOG (1,568-dim), and MobileNetV2 (1,280-dim) vectors are concatenated into a single 2,852-dimensional representation [5]. This late fusion strategy allows the classifier to simultaneously exploit pixel-level texture statistics, structural edge orientation patterns, and hierarchical semantic features. Fig. 5 illustrates the fusion process.

F. Normalization

The following normalization strategies were applied to eliminate scale bias [9]:

- **Z-score Standardization**: $z = (x - \mu)/\sigma$
- **Min-Max Normalization**: $x' = (x - \min)/(\max - \min)$
- **GLCM Normalization**: $P(i,j) = c(i,j)/\sum_{i,j} c(i,j)$
- **HOG Block L2-Normalization**: $v' = v/\sqrt{\|v\|_2^2 + \epsilon^2}$

Z-score standardization via `StandardScaler` was applied to the final fused vector before LightGBM training and inference.

G. Classification using LightGBM

LightGBM was selected due to its efficiency on high-dimensional structured tabular data [8]. It uses a leaf-wise tree growth strategy and histogram-based gradient boosting. The classifier was configured with 300 estimators and a learning rate of 0.05.

IV. Experimental Setup

A. Hardware and Software Environment

All experiments were conducted on an AMD Ryzen 5 system with 8 GB RAM using Python 3.13. Key libraries included TensorFlow/Keras for MobileNetV2, Scikit-image for GLCM/HOG, and LightGBM for classification. Table IV summarises the full configuration.

TABLE IV
Experimental Setup and Classifier Configuration

Component	Configuration
Processor	AMD Ryzen 5
RAM	8 GB
Language	Python 3.13
Deep learning	TensorFlow / Keras
Feature extraction	Scikit-image
Classifier	LightGBM
Estimators	300
Learning rate	0.05
Train/Test split	80% / 20% (stratified)
Backend	FastAPI + Uvicorn
Frontend	React + TailwindCSS
Model serialization	joblib (.pkl)

B. Dataset and Feature Configuration

The multi-modal strategy employed on PlantVillage [7]:

1) **Deep Features:** MobileNetV2 (ImageNet pretrained) $\rightarrow$ 1,280-dim GAP vector.
2) **Texture Features:** GLCM (contrast, homogeneity, energy, correlation, ASM) from 256×256 grayscale images.
3) **Shape Features:** HOG with 9 orientations, 8×8 px/cell, 2×2 cells/block.
4) **Fusion:** Concatenated to 2,852 dims; Z-score normalized via `StandardScaler`.

V. Results and Discussion

The proposed system was evaluated on 40,612 test samples across 17 disease classes. The full three-way fusion achieved an overall accuracy of **96.17%**, with precision, recall, and F1-score each at 0.96.

Table V presents the ablation study. Fig. 6 visualizes the accuracy progression across feature combinations.

TABLE V
Ablation Study: Accuracy by Feature Combination

Feature Combination	Accuracy (%)
GLCM only	26.74
HOG only	48.05
MobileNetV2 only	85.77
MobileNetV2 + GLCM	87.43
MobileNetV2 + HOG	88.49
GLCM + HOG + MobileNetV2 (Proposed)	**96.17**

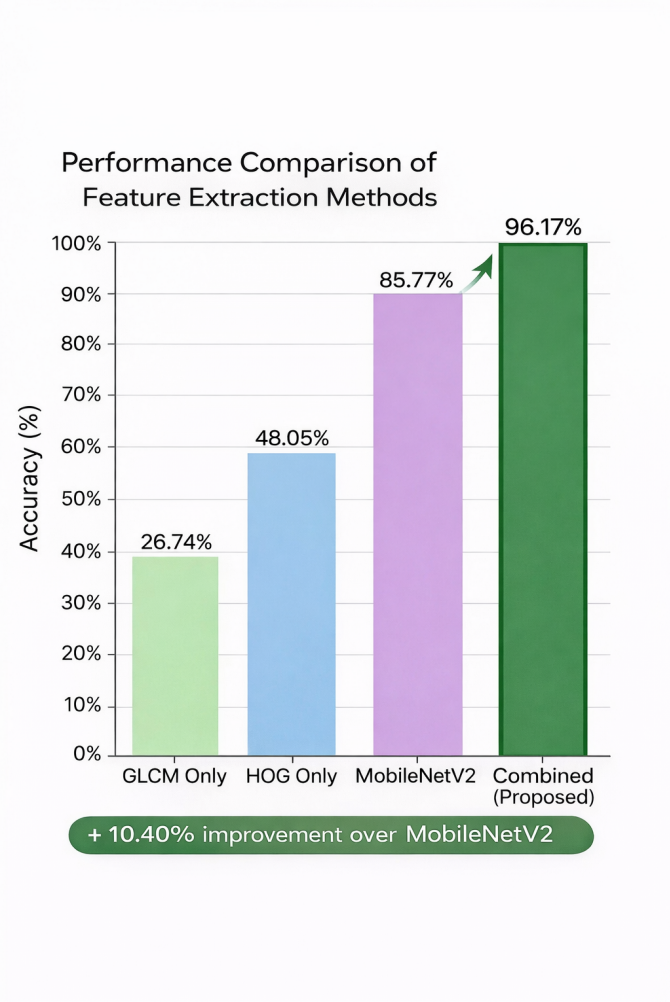


Fig. 6. Performance comparison across feature combinations, highlighting performance gain over baseline models. The proposed full fusion (GLCM + HOG + MobileNetV2) achieves 96.17%, a 10.40 percentage point improvement over MobileNetV2 alone.

Table VI shows per-class precision, recall, and F1-score for all 17 disease categories.

TABLE VI
Per-Class Performance Metrics

Disease Class	Prec.	Rec.	F1
Apple Scab	1.00	1.00	1.00
Apple Healthy	1.00	1.00	1.00
Pepper Bell Bacterial Spot	0.78	0.78	0.78
Pepper Bell Healthy	0.97	0.97	0.97
Potato Early Blight	0.99	0.99	0.99
Potato Late Blight	0.96	0.96	0.96
Potato Healthy	1.00	1.00	1.00
Tomato Bacterial Spot	0.97	0.97	0.97
Tomato Early Blight	0.90	0.90	0.90
Tomato Late Blight	0.98	0.98	0.98
Tomato Leaf Mold	0.95	0.95	0.95
Tomato Septoria Leaf Spot	0.97	0.97	0.97
Tomato Spider Mites	0.96	0.96	0.96
Tomato Target Spot	0.95	0.95	0.95
Tomato YL Curl Virus	1.00	1.00	1.00
Tomato Mosaic Virus	0.92	0.92	0.92
Tomato Healthy	0.95	0.95	0.95
Macro Average	**0.96**	**0.96**	**0.96**

The normalized confusion matrix is shown in Fig. 7. Strong diagonal dominance confirms classification reliability. Pepper Bell Bacterial Spot (0.78) shows the most off-diagonal confusion due to visual similarity with other bacterial lesion patterns.

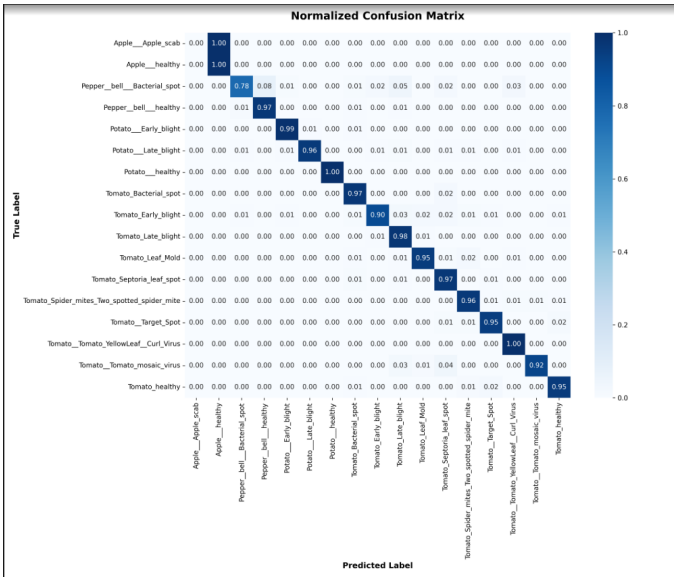


Fig. 7. Normalized confusion matrix across 17 disease classes. Diagonal values represent per-class recall. Most classes exceed 0.90, with Pepper Bell Bacterial Spot (0.78) being the most challenging.

The full three-way fusion yields a 10.40 percentage point improvement over MobileNetV2 alone (85.77%), confirming that complementary handcrafted features meaningfully augment deep representations. The proposed method also outperforms existing hybrid approaches reported in the literature by approximately 7–9%, demonstrating the advantage of three-

way fusion over two-way combinations. Backend inference latency averaged **0.42 seconds per image**, enabling real-time field deployment. Table VII summarises the runtime breakdown.

TABLE VII
INFERENCE TIME BREAKDOWN PER IMAGE

Stage	Time (s)
HSV Segmentation	0.03
GLCM Feature Extraction	0.02
HOG Feature Extraction	0.05
MobileNetV2 Deep Features	0.28
Feature Fusion & Normalization	0.01
LightGBM Classification	0.03
Total (avg.)	**0.42**

VI. CONCLUSION

This paper presented a hybrid feature fusion framework for automated crop disease detection, combining GLCM texture features, HOG structural features, and MobileNetV2 deep semantic features into a unified 2,852-dimensional representation classified by LightGBM. The proposed system, deployed as a full-stack web application (React + FastAPI), achieved an overall accuracy of 96.17% across 17 disease classes, with precision, recall, and F1-score all at 0.96.

The ablation study confirmed that each modality contributes uniquely: GLCM captures pixel-level texture variation, HOG encodes structural edge patterns, and MobileNetV2 provides hierarchical semantic features. Their fusion substantially outperforms any individual component and surpasses existing hybrid models in the literature. LightGBM proved to be an effective and efficient classifier for high-dimensional fused feature vectors, delivering 0.42-second average inference latency suitable for practical agricultural deployment.

The work successfully bridges the gap between research-oriented deep learning and practical real-world deployment for farmers and agricultural extension workers. Future work will validate the model on real-world field datasets to assess generalization.

Limitations: Despite strong results, the PlantVillage dataset predominantly contains images captured under controlled laboratory conditions with clean, uniform backgrounds. While HSV-based segmentation partially mitigates this, performance may degrade on real-field photographs with complex lighting, overlapping leaves, soil residue, or motion blur. The current model covers 17 of 38 disease classes; generalization to the remaining classes has not been evaluated. The 0.42-second inference time may be a constraint for low-power edge deployments. Future work will evaluate the model on real field images and will investigate model compression via quantization and pruning to improve edge deployment feasibility.

VII. FUTURE SCOPE

Several directions can extend and improve the proposed system in both the short and long term.

A. Short-Term Improvements

Performance on low-accuracy classes such as Pepper Bell Bacterial Spot can be improved through data augmentation strategies including random cropping, color jitter, and GAN-based synthetic sample generation. YOLO-based spatial localization can be integrated to provide bounding-box-level disease detection beyond image-level classification. An offline mobile application can extend accessibility to rural areas with limited internet connectivity, and multilingual interface support will broaden usability across India's diverse farming communities.

B. Long-Term Directions

Disease severity grading (early, moderate, severe) can be added as an additional output dimension for more precise treatment recommendations. Drone-based aerial image capture can enable large-scale field monitoring beyond individual leaf diagnosis. Temporal tracking of disease progression across successive images of the same plant can provide predictive warnings before visible damage becomes severe. Finally, the pipeline can be extended to pest detection, soil health analysis, and nutrient deficiency classification, creating a comprehensive crop health management platform.

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